1 Angles, Rotation, and the Unit Circle

Angles provide a description of rotation. On the unit circle, this description determines a point whose coordinates define the sine and cosine functions.

A rotation can be specified by its angle, independently of the size of the circle on which it acts. The relation between angle and position leads to trigonometric coordinates and to models of periodic motion.

Learning objectives. The principal objectives are to describe signed rotation, convert between degrees and radians, calculate arc length and angular speed, determine exact unit-circle coordinates, and relate uniform circular motion to periodic coordinate functions.

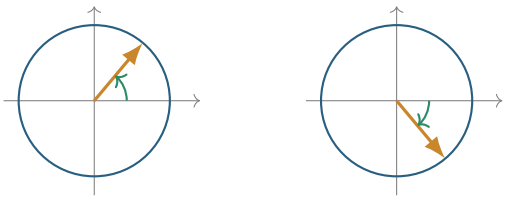
1.1 Angles measure turning

An angle records a rotation from one ray to another. Counterclockwise rotations are positive, and clockwise rotations are negative. A rotation can exceed one full revolution. For example, 450° and 90° have the same terminal ray.

Definition 1.1. Two angles are *coterminal* if they have the same initial and terminal rays. They differ by an integer multiple of 360°, or 2π radians.

Example 1.1. A positive coterminal angle for -50° is $-50^\circ + 360^\circ = 310^\circ$.

In standard position, the initial ray coincides with the positive horizontal axis. Positive angles correspond to counterclockwise rotation and negative angles to clockwise rotation. The signed angle specifies the accumulated rotation, whereas the terminal ray specifies only the final direction. Thus 30°, 390°, and -330° point along the same terminal ray, although their rotations are different.



Worked example: multiple turns. Find an angle between 0° and 360° coterminal with -770°. Adding 360° twice gives -50°; adding it once more gives 310°. The resulting angle lies in $[0^\circ, 360^\circ)$ and has the same terminal ray.

Try it. Sketch -120°, 240°, and 600°. Which terminal rays coincide? Justify the result using coterminal angles.

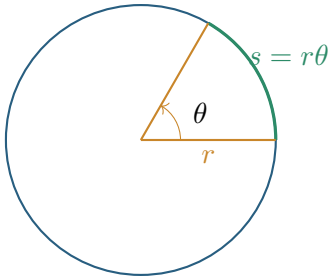
1.2 Radian measure

For a circle of radius r and an arc of length s subtended by a central angle, the magnitude of the angle in radians is s/r. For an oriented arc, the signed arc length determines the signed angle. A full circumference is $2\pi r$, so one revolution measures 2π radians:

$$\boxed{\theta = \frac{s}{r}}, \qquad 180^\circ = \pi \text{ radians}.$$

Example 1.2. Convert 150° to radians: $150(\pi/180) = 5\pi/6$. Convert $7\pi/4$ radians to degrees: $(7\pi/4)(180/\pi) = 315^\circ$.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.



Arc length and central angle

For a fixed central angle, the intercepted arc length is proportional to the radius. Consequently, the ratio s/r does not depend on the size of the circle.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	$1/12$
45°	$\pi/4$	$1/8$
60°	$\pi/3$	$1/6$
90°	$\pi/2$	$1/4$
180°	π	$1/2$
270°	$3\pi/2$	$3/4$
360°	2π	1

The conversion $180^\circ = \pi$ radians determines every entry in the table. In particular, $60^\circ = \pi/3$ radians.

Example 1.3. Convert a negative angle: $-210^\circ \cdot \pi/180^\circ = -7\pi/6$. Its terminal ray coincides with $5\pi/6$, because adding 2π to $-7\pi/6$ gives $5\pi/6$.

1.3 Arc length and angular speed

For signed radian measure θ, the directed arc length is $s = r\theta$; the corresponding distance traveled is $r|\theta|$. Over a time interval of length t, the average angular velocity is $\omega = \theta/t$ and the average tangential speed is $v = r|\omega|$. For uniform rotation, these quantities are constant.

Example 1.4. A wheel of radius 0.4 m turns through $3\pi/2$ radians. A point on its rim travels $s = 0.4(3\pi/2) = 0.6\pi$ m.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Worked example: a carousel. A carousel makes one revolution every 24 seconds. A child sits 3 meters from the axis, while another sits 5 meters from the axis. Both riders have the same angular speed,

$$\omega = \frac{2\pi}{24} = \frac{\pi}{12} \text{ rad/s}.$$

Their tangential speeds differ:

$$v_1 = 3\frac{\pi}{12} = \frac{\pi}{4} \text{ m/s}, \qquad v_2 = 5\frac{\pi}{12} = \frac{5\pi}{12} \text{ m/s}.$$

Tangential speed depends on the distance from the axis, whereas angular speed does not.

The arc-length formula requires radian measure. For example, a quarter-turn is $\pi/2$ radians, so its arc length is $\pi r/2$.

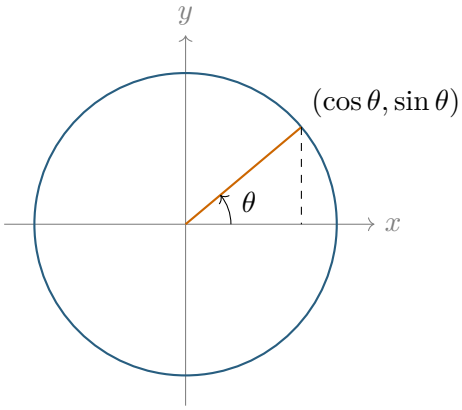
Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

1.4 Coordinates on the unit circle

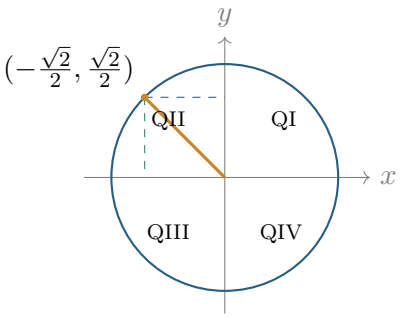
Take a circle of radius 1 centered at the origin. The point at angle θ, measured from the positive x-axis, has coordinates

$$\boxed{(x, y) = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate and sine is the vertical coordinate. Because the point lies on $x^2 + y^2 = 1$, we get $\cos^2\theta + \sin^2\theta = 1$.



Rotating the point $(1, 0)$ through θ produces $(\cos\theta, \sin\theta)$. Both coordinates are signed quantities; their signs depend on the quadrant containing the terminal ray.



Worked example: a point in Quadrant III. At $225° = 5\pi/4$, the reference angle is $\pi/4$. The two coordinates have equal absolute value, but both are negative:

$$(\cos(5\pi/4), \sin(5\pi/4)) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

The reference triangle determines the coordinate magnitudes; the quadrant determines their signs.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

1.5 Reference angles and periodicity

Special angles arise from right triangles:

θ	$\cos\theta$	$\sin\theta$
0	1	0
$\pi/6$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1

The reference angle supplies the absolute values in every quadrant; the quadrant gives the signs.

Example 1.5. At $5\pi/6$, the reference angle is $\pi/6$ and the point is in quadrant II, so $\cos(5\pi/6) = -\sqrt{3}/2$ and $\sin(5\pi/6) = 1/2$.

After one full revolution the point returns to its previous position:

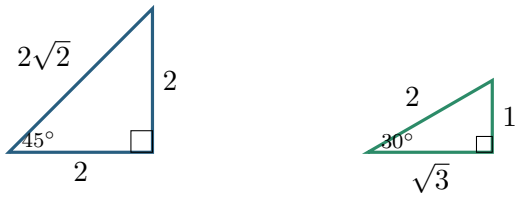
$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This geometric periodicity will become the foundation of sinusoidal graphs.

1.6 Where exact coordinates come from

The standard exact values follow from the geometry of two special right triangles.

An isosceles right triangle with legs 1 has hypotenuse $\sqrt{2}$. Dividing both legs by $\sqrt{2}$ yields the point $(\sqrt{2}/2, \sqrt{2}/2)$ at 45°. Likewise, a 30°–60°–90° triangle has side lengths in the ratio $1 : \sqrt{3} : 2$. Dividing by its hypotenuse produces the coordinates $(\sqrt{3}/2, 1/2)$ for 30° and $(1/2, \sqrt{3}/2)$ for 60°.



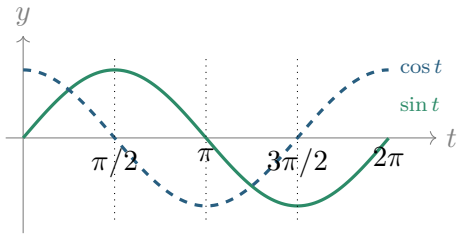
On the unit circle, the hypotenuse is normalized to length 1, and the remaining side lengths give the absolute coordinate values.

1.7 From the circle to a wave

For uniform rotation with angular velocity ω and initial angle zero, let

$$\theta(t) = \omega t, \qquad x(t) = \cos(\omega t), \quad y(t) = \sin(\omega t).$$

The point traces a circle in the coordinate plane; each coordinate, regarded as a function of time, oscillates periodically.



These graphs represent coordinate values as functions of time, rather than the circular trajectory in the xy-plane.

1.8 Example: a Ferris-wheel model

A Ferris wheel has radius 8 meters, its center is 10 meters above ground, and a passenger starts at the rightmost point. Suppose it rotates counterclockwise once every 40 seconds. Its angular speed is $\omega = 2\pi/40 = \pi/20$ radians per second, and the passenger's position relative to the ground is

$$\boxed{x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad y(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).}$$

At $t = 0$ the passenger is at $(8, 10)$. At $t = 10$ seconds the wheel has turned through $\pi/2$, so the passenger is at $(0, 18)$. At $t = 20$ seconds the passenger is at $(-8, 10)$. The coordinate equations determine the passenger's position at any time.

The radius determines the amplitude of both coordinate functions. The center height shifts the vertical coordinate, and the period determines the angular velocity.

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.9 Exercises

1. Convert $-135°$ and $300°$ to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.

11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.

12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.

13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

The unit-circle coordinates also determine tangent, cotangent, secant, and cosecant, which are developed in the following chapter.